

BRAC UNIVERSITY
Department of Computer Science and Engineering

Examination: Quiz - 2
Duration: 25 minutes

Semester: Summer 2026
Full Marks: 20

CSE 340: Computer Architecture

Name:	ID:	Section:
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- 1. Convert** the given RISC-V assembly instructions into their corresponding machine code. Show the hex equivalent of the representation.

i. LBU x6, 40(x22) [Opcode field value is (3)₁₀] [4.5+0.5]

Machine Code (binary)	00000010100010110100001100000011
Machine Code (Hex)	0x028B4303

ii. SD x7, 8(x21) [Opcode field value is (35)₁₀] [4.5+0.5]

Machine Code (binary)	00000000011110101011010000100011
Machine Code (Hex)	0x007AB423

Note:

Refer to the following table to find the funct3 value of the Load and Store instruction.

	Data Size	Funct3		Data Size	Funct3
Moving data from memory to register. (Signed)	8 bits	000	Moving data from register to Memory.	8 bits	000
	16 bits	001		16 bits	001
	32 bits	010		32 bits	010
	64 bits	011		64 bits	011
Moving data from memory to register. (Unsigned)	8 bits	100			
	16 bits	101			
	32 bits	110			

- 2.** For the below statements, identify if true or false. For true just write true. For any false statement, write false and the correct statement or the reason why the statement is false.

a. RISC-V follows big endian approach while storing data. [1]

Answer: False. Little endian.

b. Size of the rs field is 5 bits in the RISC-V version that we follow. [1]

Answer: True

3. B is a byte **word** array with 90 indexes. How many memory locations are needed to store the array B in the RISC-V system? [2]

Answer: $90 \times 8 = 720 \text{ bits} / 8 = 90 \text{ (byte)}$

$90 \times 32 = 2880 \text{ bits} / 8 = 360 \text{ (word)}$

The question was supposed to have either byte or word. Hence, I gave full marks to both pirates.

4. C is a double word array. The starting address of C[15] is stored in register x25. Write necessary RISC-V code to calculate the base address of array C in register x26. [3]

Answer:

Target = Base + Offset $\Rightarrow$ Base = Target - Offset, Target is C[15] = x25; Offset = ?

Offset = index $\times$ 8 (double word array) = $15 \times 8 = 120$

Base = Target - 120

Addi x7, x0, 15 # x7 = 15

Slli x7, x7, 3 # x7 = x7 * 8

Sub x26, x25, x7

Or,

Addi x26, x25, -120

5. Given the contents of register X₁₉

0000 0000 0000 0000 0000 0000 0000 0000 1011 0110 1101 1110 0011 1010 1011 0100

For further analysis, we need to copy only the bits at the 4th, 7th, 9th, 12th, 20th, 25th and 28th positions.

- a) Construct a mask to do so. [1]

Answer: Mask =

0000 0000 0000 0000 0000 0000 0000 0000 0001 0010 0001 0000 0001 0010 1001 0000

- b) Write the name of the instruction that can be used to copy these bits using the prepared mask. [2]

Answer: And